

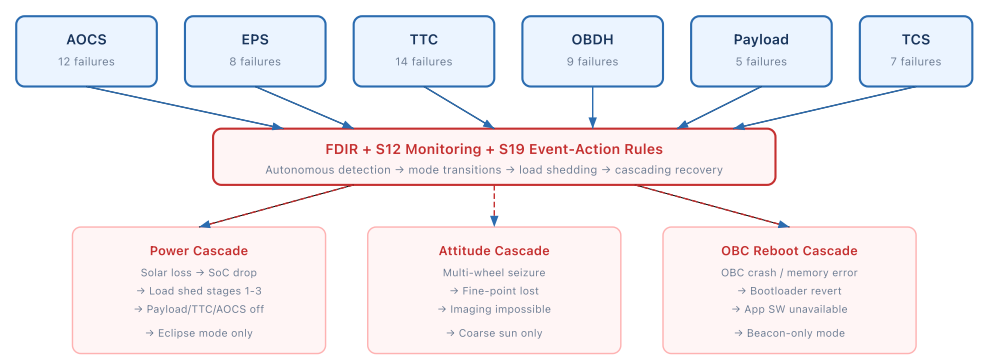
EOSAT-1 Failure Modes Catalog

Injectable Failure Modes, Effects, and Cascading Chains

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1. System Overview

The EOSAT-1 simulator implements **42 injectable failure modes** across 6 subsystems, with autonomous fault detection (FDIR), on-board monitoring (S12), and event-action rules (S19) that create realistic cascading failure chains.



2. AOCS Failures 12

File: smo_simulator/models/aocs_basic.py

Failure	Parameters	Immediate Effect	Cascading
rw_bearing	wheel (0-3), magnitude	Bearing degradation; wheel disabled if mag ≥ 0.95	If <3 wheels → COARSE_SUN mode
rw_seizure	wheel (0-3)	Wheel speed → 0, disabled	Multi-wheel → FINE_POINT lost → imaging impossible
gyro_bias	axis (0-2), bias (deg/s)	Constant rate bias on gyro	Attitude error grows → S19 rule 2004 safe mode

st_blind	unit, magnitude	Tracker status → BLIND	FINE_POINT → NOMINAL fallback; random gyro bias injected
st_failure	unit (1-2)	Tracker status → FAILED, num_stars=0	Permanent; combined with CSS loss → DETUMBLE only
css_failure	—	css_valid = False	Sun vector lost; COARSE_SUN mode blocked
mag_failure	—	mag_valid = False	Desaturation degraded (no magnetic field reference)
mtq_failure	axis (x/y/z)	Single MTQ axis disabled	Desaturation effectiveness reduced ~33%
mag_a_fail	—	MAG-A failed	Switch to MAG- B; both → mag_failure cascade
mag_b_fail	—	MAG-B failed	Switch to MAG- A; both → mag_failure cascade
css_head_fail	face (px/mx/py/my/pz/mz)	Single CSS head disabled	Reduced sun vector accuracy; multiple → css_failure
multi_wheel_failure	wheels (list)	2+ wheels disabled simultaneously	Forces COARSE_SUN; eliminates fine- pointing

3. EPS Failures 8

File: smo_simulator/models/eps_basic.py

Failure	Parameters	Immediate Effect	Cascading
solar_array_partial	array (A/B), magnitude	Array power reduced	SoC drops → may trigger load shedding

bat_cell	magnitude	Battery cell failure flag	Capacity loss → faster SoC decline under load
bus_short	magnitude	Bus short flag	Overcurrent protection trips multiple lines
overcurrent	line_index (0-7), magnitude	OC injection on power line	Line trips → subsystem power loss
undervoltage	magnitude	SoC reduced by mag×10%	Progressive load shedding: Payload→TTC→AOCS
overvoltage	magnitude	OV threshold lowered	May trigger widespread power line trips
solar_panel_loss	face, magnitude	Single body panel degraded	Power reduction; compounds with other panels
solar_array_total_loss	—	All panels + arrays → 0 power	Battery-only → catastrophic load shedding → eclipse mode

4. TTC Failures 14

File: smc_simulator/models/ttc_basic.py

Failure	Parameters	Immediate Effect	Cascading
primary_failure	magnitude	Primary transponder dead	Link lost until redundant switchover
redundant_failure	magnitude	Redundant transponder dead	Combined → total comms loss
high_ber	magnitude (dB offset)	Eb/N0 reduced	Frame sync loss risk; S19 rule 4003 TX power increase
pa_overheat	magnitude (heat watts)	PA temperature rising	Auto-shutdown at 70°C → downlink lost
uplink_loss	—	carrier_lock = False	No command acceptance; spacecraft autonomous only

receiver_degrade	magnitude (dB NF)	Noise figure increase	Lock acquisition delays; eventual uplink loss
antenna_deploy_failed	—	Antenna stowed, sensor jammed	20-30 dB link budget loss; effective comms loss
<i>Ground segment failures (7): gs_lna_degradation, gs_antenna_mispoint, gs_feed_loss, gs_rfi_interference, gs_hpa_degradation, gs_ref_osc_drift, gs_tracking_loss — each injects dB loss into ground link budget; cumulative; no spacecraft-side cascade</i>			

5. OBDH Failures 9

File: smosimulator/models/obdh_basic.py

Failure	Parameters	Immediate Effect	Cascading
watchdog_reset	—	Arms watchdog at period	Timeout → reboot → bootloader → phase regression
memory_errors	count	EDAC error count increment	S19 rule 5003 → memory scrub
cpu_spike	load (%)	CPU baseline raised	S19 rule 5004 → safe mode if threshold exceeded
obc_crash	—	Immediate reboot	Bootloader → phase regression → beacon-only HK
bus_failure	bus (A/B)	CAN bus status → FAILED	S19 rule 5002 → switch to redundant bus
boot_image_corrupt	—	CRC check will fail	Reboot → stuck in bootloader indefinitely
memory_corruption	count	Uncorrectable EDAC + reboot	Same as obc_crash + memory error events
memory_segment_fail	segment (0-3)	Segment marked bad	Reduced storage capacity; affects payload imaging
stuck_in_bootloader	—	Boot inhibit + image corrupt	Locked in bootloader; operator recovery required

6. Payload Failures 5

File: smo_simulator/models/payload_basic.py

Failure	Parameters	Immediate Effect	Cascading
cooler_failure	magnitude	FPA cooler disabled	FPA temp rises → S19 rule 6002 → payload off
fpa_degraded	magnitude	Error probability 2×	Image quality loss; combined with cooler → severe
image_corrupt	count	Next N images corrupted	Data loss; storage waste
memory_segment_fail	segment	Segment marked bad	Storage capacity reduced → S19 rule 6001 if full
ccd_line_dropout	magnitude	Error probability 3×	Combined with fpa_degraded → most images unusable

7. TCS Failures 7

File: smo_simulator/models/tcs_basic.py

Failure	Parameters	Immediate Effect	Cascading
heater_failure	circuit, magnitude	Heater disabled	Zone cools; battery heater → EPS capacity loss
cooler_failure	magnitude	FPA cooler disabled	FPA warms → payload off cascade
obc_thermal	heat_w	OBC heat injection	OBC + battery overtemp → S19 rules 3002/3003
sensor_drift	zone, magnitude	Temp sensor offset	False alarms; operator confusion
heater_stuck_on	circuit	Heater always ON	Zone overheats → thermal runaway possible
heater_open_circuit	circuit	Heater draws no power	Zone cools uncontrollably
temp_anomaly	zone, offset_c	Immediate temp offset	False overtemp alarm → cascading subsystem shutdown

8. Cascading Failure Chains

Six critical cascading chains have been identified where a single failure propagates across multiple subsystems:

Chain 1: Solar Array Total Loss → Catastrophic Load Shedding

```
solar_array_total_loss
→ power_gen = 0W (no solar input)
→ bat_soc drops ~2%/min under load
→ SoC < 30%: LOAD_SHED_STAGE_1 → Payload OFF
→ SoC < 20%: LOAD_SHED_STAGE_2 → TTC TX OFF, AOCS wheels
OFF
→ SoC < 10%: LOAD_SHED_STAGE_3 → All non-essential OFF
→ AOCS forced to MODE_ECLIPSE (gyro-only propagation)
→ Spacecraft in safe/beacon mode until solar recovery
```

Chain 2: Multi-Wheel Seizure → Attitude Control Loss

```
rw_seizure[wheel=0] + rw_seizure[wheel=1]
→ active_wheels < 3
→ MODE_FINE_POINT → MODE_NOMINAL (fallback)
→ MODE_NOMINAL → MODE_COARSE_SUN (if <3 wheels)
→ Fine-resolution imaging impossible
→ Payload mode forced to STANDBY
→ Mission science capability lost
```

Chain 3: OBC Crash → Bootloader Revert

```
obc_crash OR memory_corruption
→ _reboot(REBOOT_WATCHDOG)
→ sw_image = SW_BOOTLOADER
→ engine._tick_spacecraft_phase detects sw_image=0
→ _enter_bootloader_mode()
→ Phase → BOOTLOADER_OPS (phase 3)
→ All HK SIDs disabled (except SID 11 beacon)
→ Application commands rejected
→ If boot_image_corrupt: STUCK indefinitely
```

Chain 4: Thermal Overtemp → Multi-Subsystem Shutdown

```
obc_thermal[30W] + cooler_failure
→ OBC temp rises → battery temp rises (thermal coupling)
→ S12 param 0x0407 (bat_temp) exceeds limit
→ S19 rule 3002 → S8 func 12 (transponder TX off)
→ Downlink lost
→ S12 param 0x0406 (obc_temp) exceeds limit
→ S19 rule 3003 → S8 func 40 (OBDH safe mode)
→ Limited operations only
```

Chain 5: Uplink Loss → Autonomous-Only Operation

```

uplink_loss OR antenna_deploy_failed
  → carrier_lock = False
  → RX_available = False → No TC acceptance
  → Operator cannot send commands
    → S12/S19 autonomous rules still function
    → Spacecraft enters safe mode autonomously
    → Recovery requires physical link restoration

```

Chain 6: Sensor Cascade → Attitude Degradation

```

st_failure[unit=1] + css_failure
  → st_valid = False, css_valid = False
  → MODE_FINE_POINT → MODE_NOMINAL (line 685-687)
  → MODE_COARSE_SUN blocked (css_valid required)
  → Gyro-only propagation; drift accumulates
  → att_error exceeds S12 threshold
    → S19 rule 2004 → AOCS safe mode
    → Fine-pointing and imaging lost

```

9. FDIR Integration

Three-Level Response:

- **Level 1 (Warning):** S5 event generated, no mode change
- **Level 2 (Alert):** Forces spacecraft to SAFE mode if currently NOMINAL
- **Level 3 (Critical):** Forces spacecraft to EMERGENCY mode immediately

Key S19 Event-Action Rules

Rule	Trigger	Action	Subsystem
1001	Bus undervoltage	Payload OFF	EPS → Payload
1002-1003	SoC critical (<20%)	FPA cooler OFF, Payload OFF	EPS → TCS/Payload
2001	Momentum saturation	Desaturate wheels	AOCS
2002-2003	ST1/ST2 failure	Switch tracker	AOCS
2004	Attitude error high	AOCS safe mode	AOCS
3001	FPA overtemp	Payload OFF	TCS → Payload
3002	Battery overtemp	TTC TX OFF	TCS → TTC
3003	OBC overtemp	OBDH safe mode	TCS → OBDH
4002	PA overtemp	PA OFF	TTC
5001	OBC reboot	Acknowledge	OBDH

5002	Excessive reboots	Switch CAN bus	OBDH
6001	Storage full	Imaging OFF	Payload

10. Autonomous Load Shedding

